BCH 4121: Tissue Biochemistry

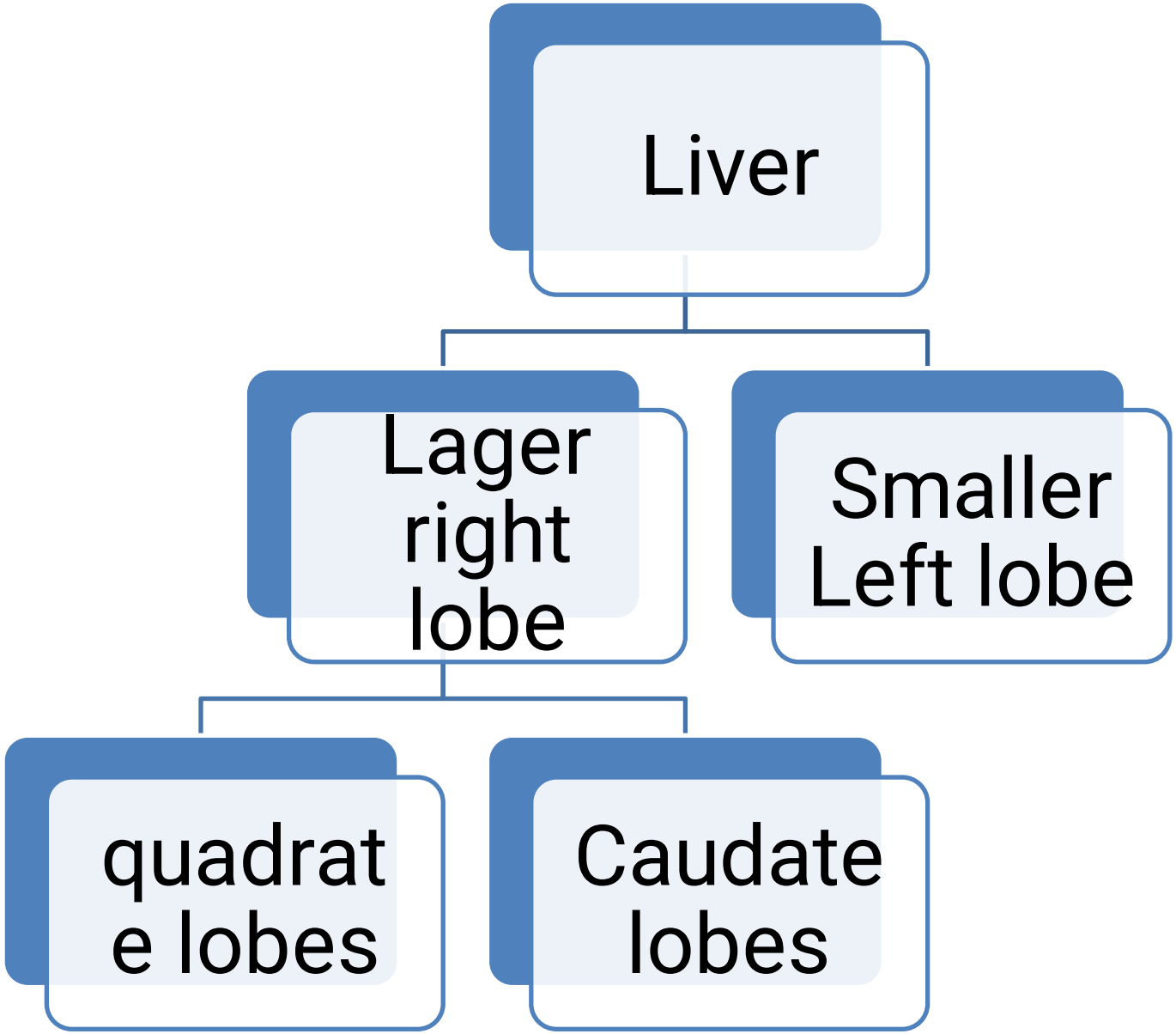
❖ Biochemistry of hepato - biliary system

by

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2023/2024 Session

➤ Properties of Liver

- ✔ Largest (1.5kg) of all visceral organs
- ✔ Representing 23% of the total body mass oxygen consumption > its %weight (25 - 30%)
- ✔ 25% of it can regenerate to whole liver.


Liver
Lager right lobe
Smaller Left lobe
quadrate lobes
Caudate lobes

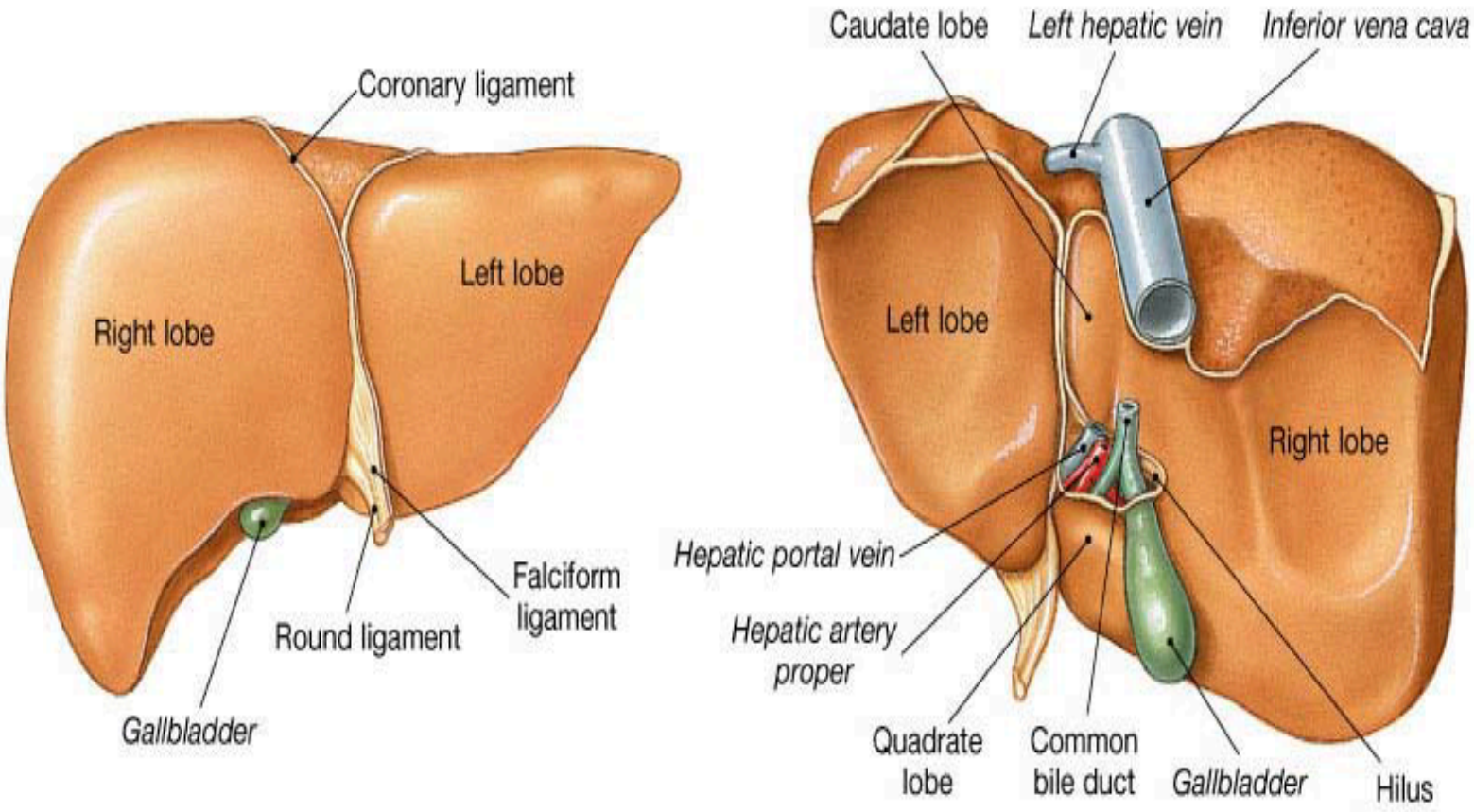


Figure: Biochemical anatomy of liver

- ✔ The falciform ligament :
  - o separates the right and left lobes anteriorly
  - o Suspends the liver from the diaphragm and anterior abdominal wall.

- ✔ Blood and nutrients supply:
  - o Hepatic artery carries oxygenated blood (20 – 30%) from the aorta

  - o Portal vein carries blood (70 – 80%) from small intestine, stomach, spleen, pancreas and large intestine.

- Total blood supply by the hepatic artery and portal vein is about 1500ml/min

- The hepatic arterial blood is 95% oxygenated (19% oxygen), portal venous is about 85% (17% oxygen) and hepatic venous blood is about 65% (13.4% oxygen)
- Portal system blockage leads to :
  - Portal hypertension that increase GIT capillary pressure with marked fluid transduction into peritoneal cavity that causes **ascites**

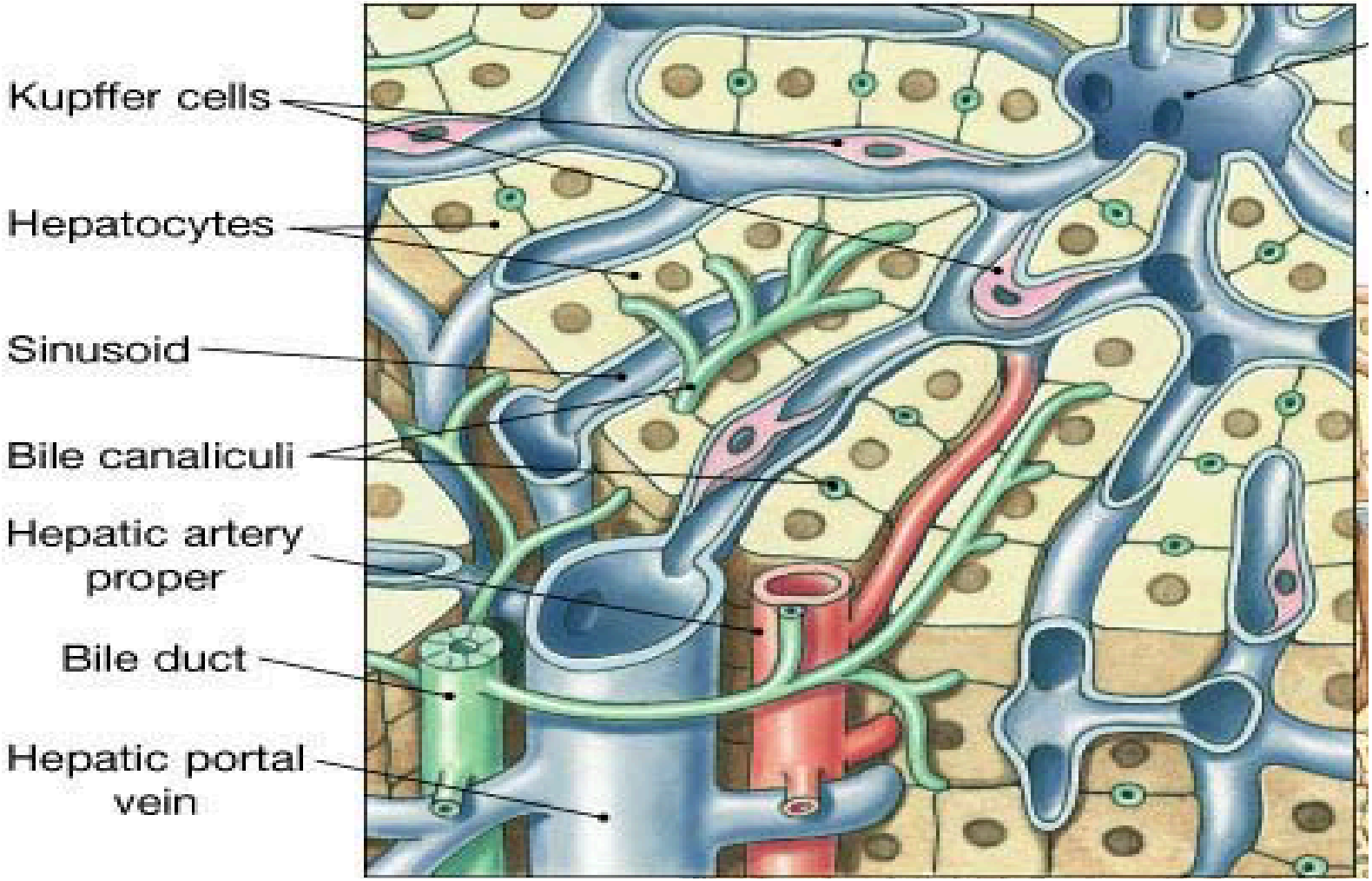


Figure: Biochemical histology of liver

- ✔ The lobes are divided to lobules
- ✔ Each lobules consists of radiating layers of cells (hepatocytes)
- ✔ The hepatocytes are blessed with biliary canaliculi into which bile is discharged
- ✔ The canaliculi join to form larger bile duct, which unite to form the hepatic duct
- ✔ The hepatic duct becomes the bile duct on joining to cystic duct from the gallbladder.

- ✔Gallbladder is a pear shape sac store under the right lobe

- ✔ The common bile duct fuses with the pancreatic duct
  before entering the duodenum.

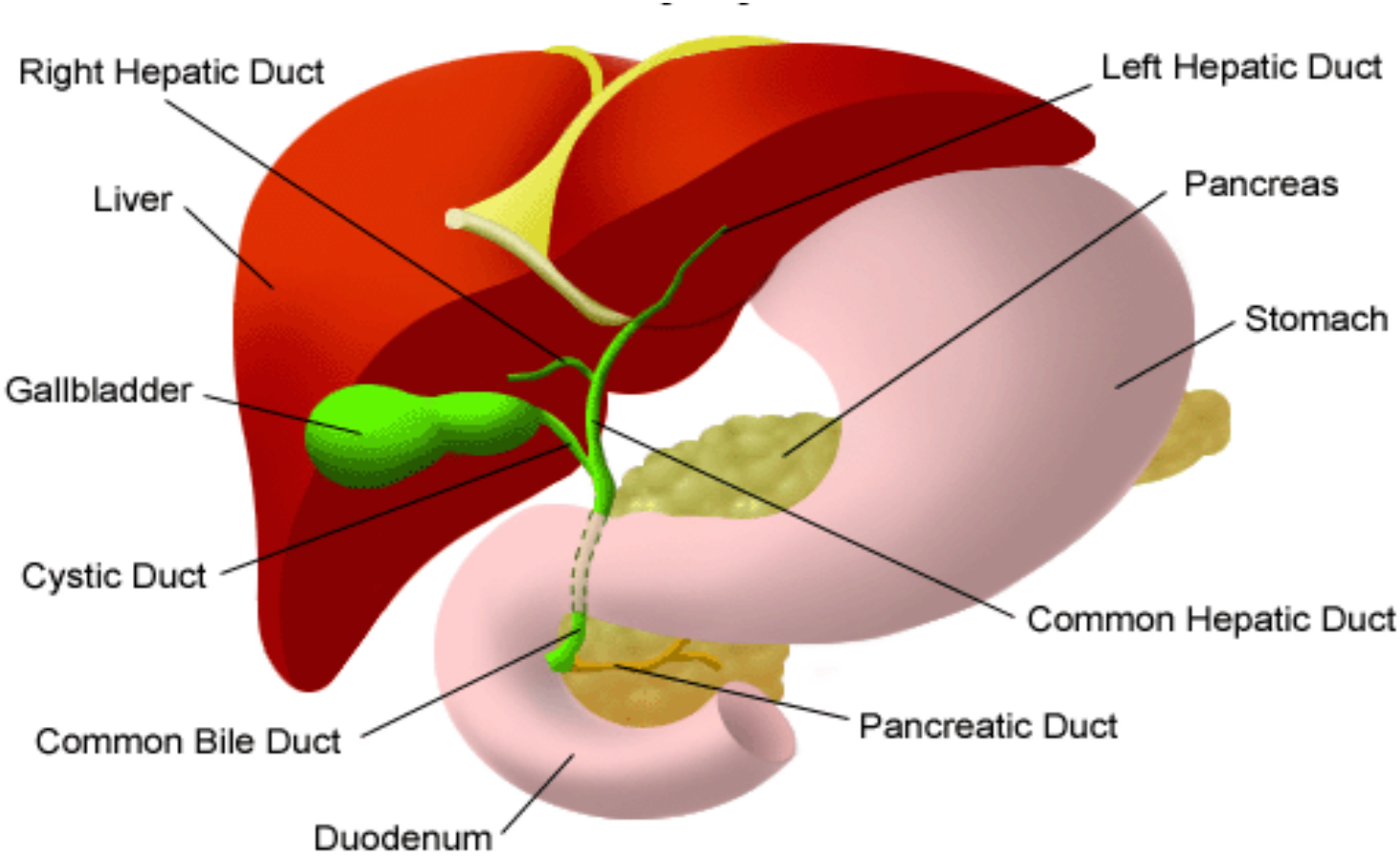


Figure: Biochemical Anatomy of hepato-biliary system

❖ FUNCTION OF LIVER

- The functional unit (hepatocytes) has over 1,000 essential enzymes to perform over 500 activities.
- **Produces and excrete bile**
  - ✔ *Excreted directly*
  - ✔ *Stored in gallbladder*
- **Perform several role in Carbohydrate metabolism**
  - ✔ gluconeogenesis
  - ✔ Glycogenolysis
  - ✔ Glycogenesis
  - ✔ Breakdown of insulin and other hormones

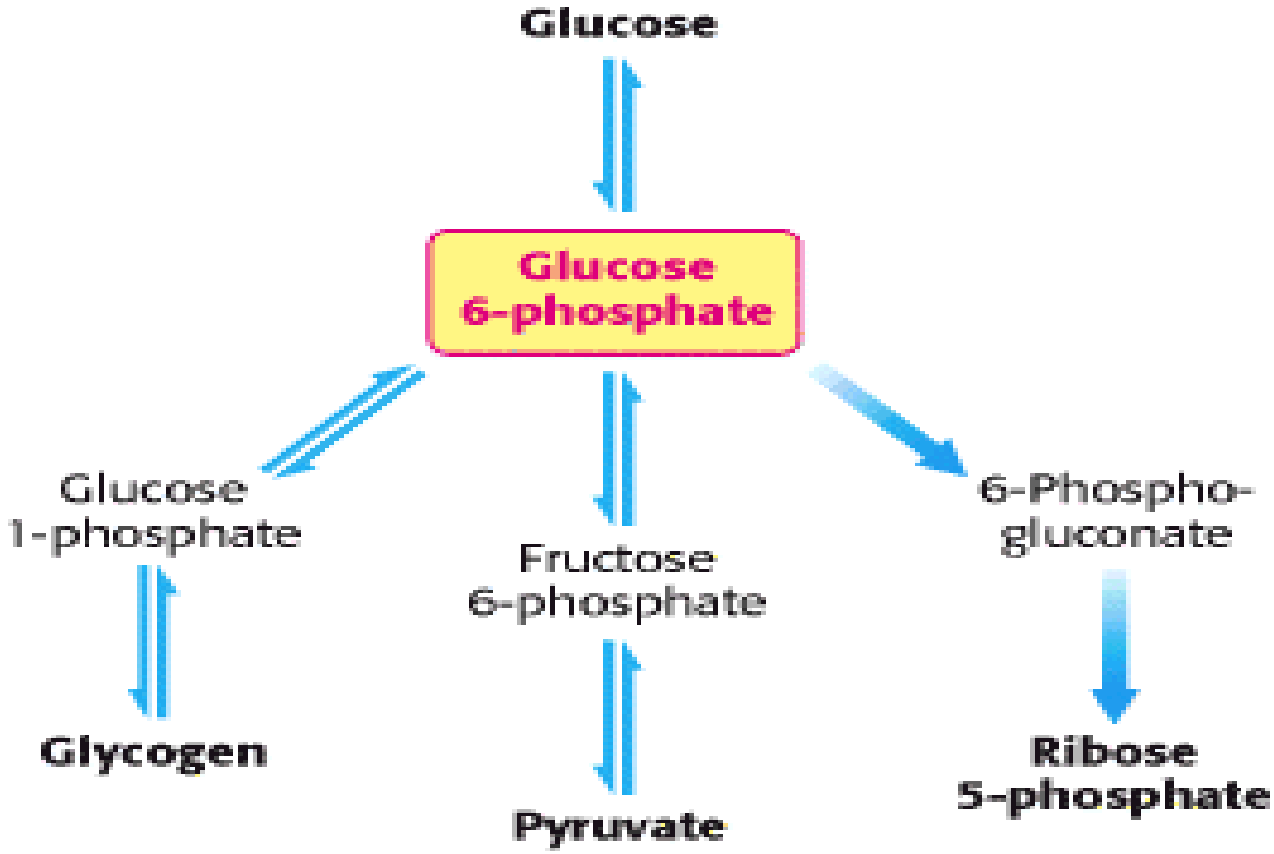
- The liver stores glycogen, converts fructose and galactose to glucose.

- It acts as buffer in the supply of glucose for fuel.

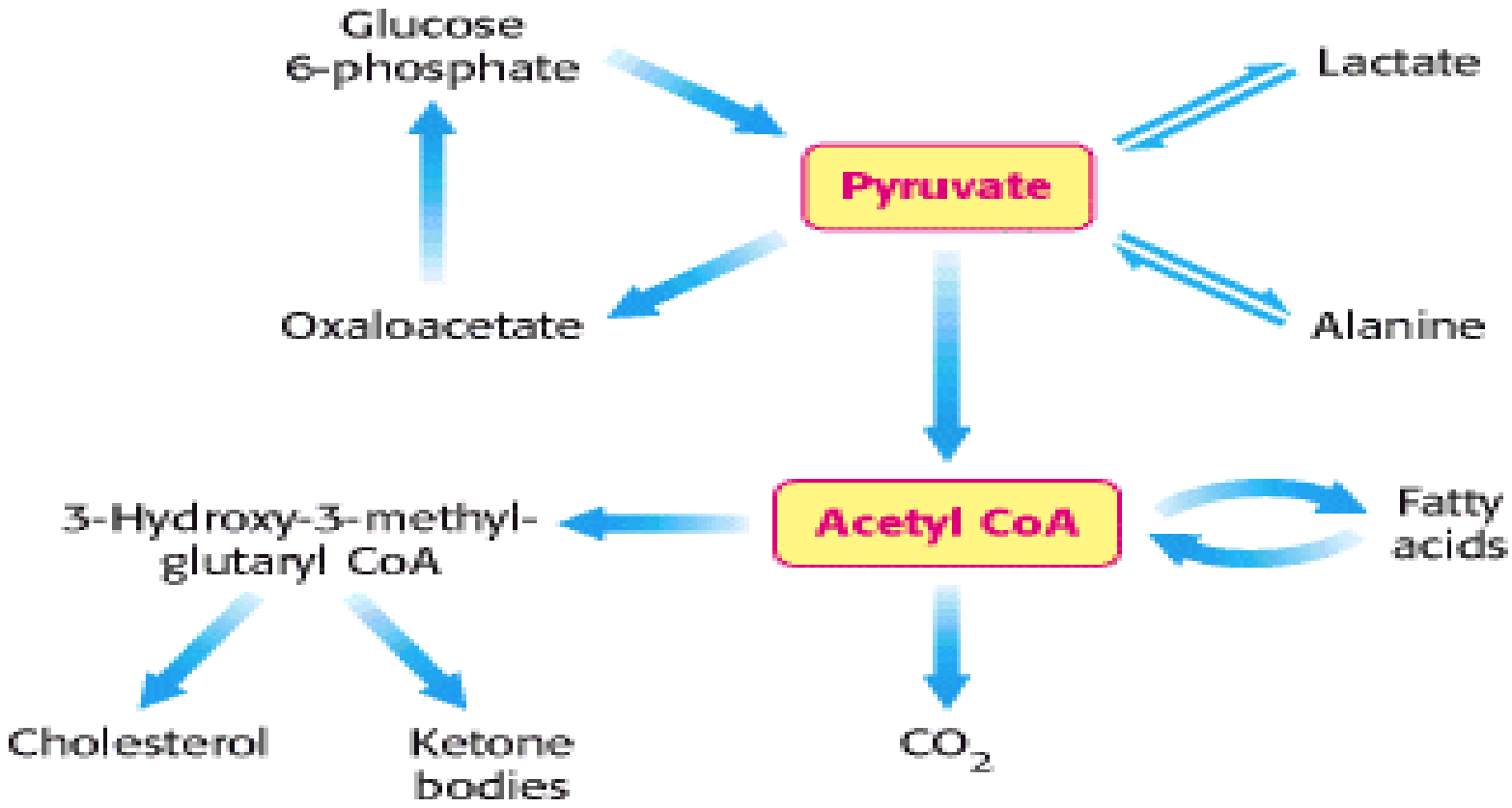
- Def iciency of enzyme (s) involves in production and breakdown of liver glycogen lead to Glycogen storage disease for e.g. Von Gierke's disease (lack of glucose – 6 – phosphatase).

Question: outline the importance and control site(s)

- It removes two-thirds of blood glucose and all other monosaccharaides.
- hexokinase and the liver-specific glucokinase convert glucose to glucose 6-phosphate.
- Fates of glucose 6- P?
- little of it used by the liver to meet its own energy needs.
- Much of the G6P is converted into glycogen [400 kcal (1700 kJ)] can be stored in this form in the liver.
- Excess G6P is metabolized to acetyl CoA, for fatty acids, cholesterol, and bile salts.



Metabolic Fates of Glucose 6-Phosphate



Major Metabolic Fates of Pyruvate and Acetyl CoA in Mammals

- The pentose phosphate pathway, another means of processing glucose 6-phosphate, supplies the NADPH for these reductive biosyntheses.
- The liver can produce glucose for release into the blood by breaking down its store of glycogen and by carrying out gluconeogenesis:

✔ *lactate and alanine from muscle*
✔ *glycerol from adipose tissue*
✔ *glucogenic amino acids from the diet.*

- Produces coagulation factors
- Storage(Glycogen,vitB12, iron, Cu^{+},
- Activates some vitamins Vit B, C, A, D, E, K

- In the f irst trimester liver is the main site of RBC later bone marrow.

- Kuffa cells are responsible for Immunological effect and reticuloendithelial system of the liver.

- These analogical active cells act as a sieve for antigen via portal system, get rid of bacteria, drug, chemicals.

- **Protein metabolism**
- ✔ Soluble plasma Fibre nactin
- ✔ Various globulins
- ✔ Liver is the only site of albumin biosynthesis
- ✔ Responsible for the generation of alpha fetoprotein before and shortly after birth.

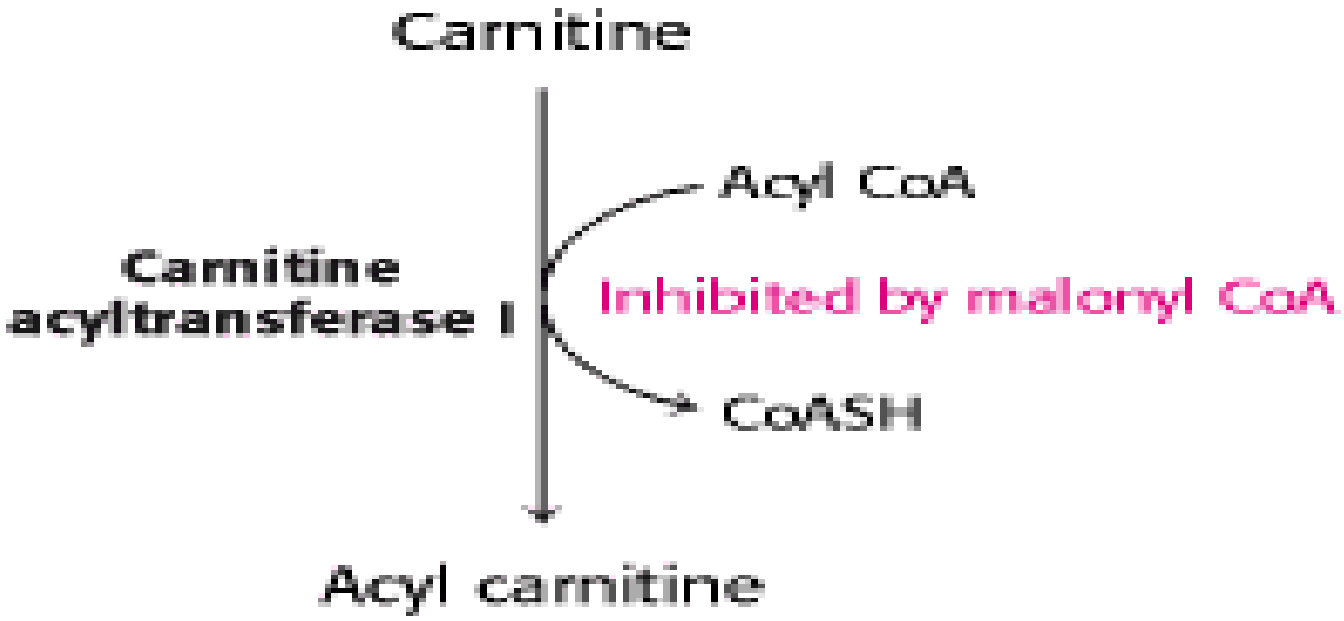
NB: in adult alpha fetoprotein is found in patient with hepatic carcinoma.

- The liver absorbs majority of dietary amino acids, leaving some in the blood for peripheral tissues.
- The priority are used for protein synthesis rather than catabolism.
- The K_m value for the aminoacyl-tRNA synthetases is lower than that of the enzymes for amino acid catabolism.
- Catabolism involves removal of nitrogen, which is subsequently processed to urea.
- It secretes 20 to 30 g of urea daily.
- The α-ketoacids are used for gluconeogenesis or fatty acid synthesis.

- Liver cannot remove nitrogen from the branch-chain amino acids (leucine, isoleucine, and valine). Transamination takes place in the muscle.
- α-Ketoacids are the liver's own fuel.
- the main role of glycolysis in the liver is to form building blocks for biosyntheses.
- it has little of the transferase needed for acetoacetate's activation to acetyl CoA. Thus, the liver eschews the fuels that it exports to muscle and the brain.

- Lipid metabolism
- Cholesterol biosynthesis
- Lipogenesis (triglyceride)
- two carbon compounds forms in the liver are burned in carbohydrate metabolic cycles
- Lipoproteins, phospholipids and steroids are synthesized in the liver.
- Proteins and carbohydrate can be converted ultimately to fats.
- 80% of the cholesterol synthesized is converted to bile acids.

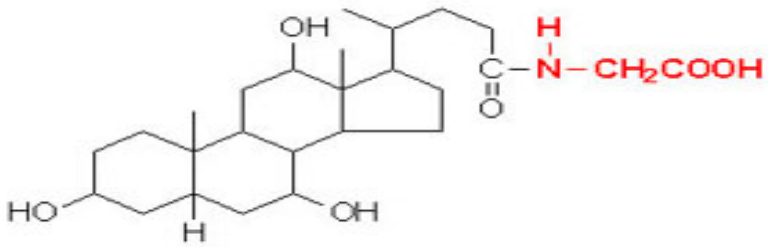
- When fuels are abundant, fatty acids derived from the diet or synthesized by the liver are esterified and secreted into the blood in the form of very low density lipoprotein.
- in the fasting state, the liver converts fatty acids into ketone bodies.
- The selection is made according to whether the fatty acids enter the mitochondrial matrix
- Recall that long-chain fatty acids traverse the inner mitochondrial membrane only if they are esterified to carnitine.



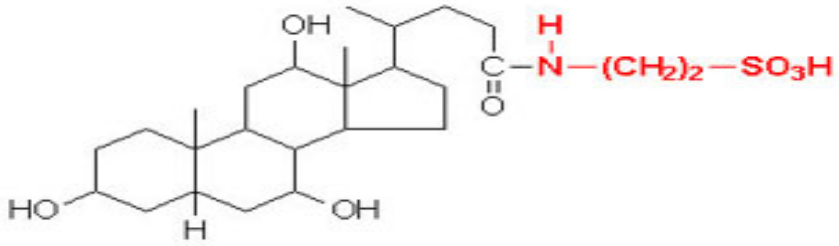
Control of Fatty Acid Degradation. Malonyl CoA inhibits fatty acid degradation by inhibiting the formation of acyl carnitine

- Carnitine acyltransferase I, is inhibited by malonyl CoA, (committed intermediate in the synthesis of fatty acids).
- *when malonyl CoA is abundant, long-chain fatty acids are prevented from entering the mitochondrial matrix, the compartment of β-oxidation and ketone-body formation.*
- *Instead, fatty acids are exported to adipose tissue for incorporation into triacylglycerols*.
- In contrast, the level of malonyl CoA is low when fuels are scarce.
- Under these conditions, fatty acids liberated from adipose tissues enter the mitochondrial matrix for conversion into ketone bodies.

- Most abundant bile acids in human bile
  - ✔ **Chenodeoxycholic acid** (45%)
  - ✔ **Cholic acid** (31%)
  - ✔ Both referred to as the **primary bile acids**.
  - ✔ Within the intestines the primary bile acids are acted upon by bacteria and converted to the



Glycocholic acid



Taurocholic acid

Structure of Conjugated cholic acids

➤ Synthesis of Bile

- ✔ In the liver carboxyl group of primary and secondary bile acids is conjugated via an amide bond to either glycine or taurine before their re-secreted into the bile canaliculi.
- ✔ These conjugation reactions respectively yield **glycoconjugates** and **tauroconjugates**
- ✔ The bile canaliculi join with the bile ductules, which then form the bile ducts.

- ✔ Bile acids are carried from the liver through these ducts to the gallbladder, where they are stored for future use.

- ✔ The ultimate fate of bile acids is secretion into the intestine, where they aid in
  - o the emulsification of dietary lipids.
  - o In the gut the glycine and taurine residues are removed
  - o the bile acids are either excreted (only a small percentage) or reabsorbed by the gut and returned to the liver

- ➤ Composition of Bile
- ✔ A yellow-green, alkaline solution containing bile salts, bile pigments, cholesterol, neutral fats, phospholipids, and electrolytes
- Bile salts are cholesterol derivatives that
- Emulsify fat
- Facilitate fat and cholesterol absorption
- solubilize, cholesterol
- ✔ Enterohepatic circulation recycles bile salts
- ✔ The chief bile pigment is bilirubin

- **Gallbladder**
- ✔ Thin-walled, green muscular sac on the ventral surface of the liver
- ✔ Stores and concentrates bile by absorbing its water and ions
- ✔ Releases bile via the cystic duct, which f bws into the bile duct

- **Regulation of bile Release**
- ✔ Acidic, fatty chyme causes the duodenum to release:
  - o Cholecystokinin(CCK) and secretin into the blood stream

- ✔ Bile salts and secretin transported in blood stimulate the liver to produce bile

- ✔ Vagal stimulation causes weak contractions of the gallbladder.

- ✔ Cholecystokinin causes:
  - o The gallbladder to contract
  - o The hepato-pancreatics phincter to relax

THEN bile enters the duodenum

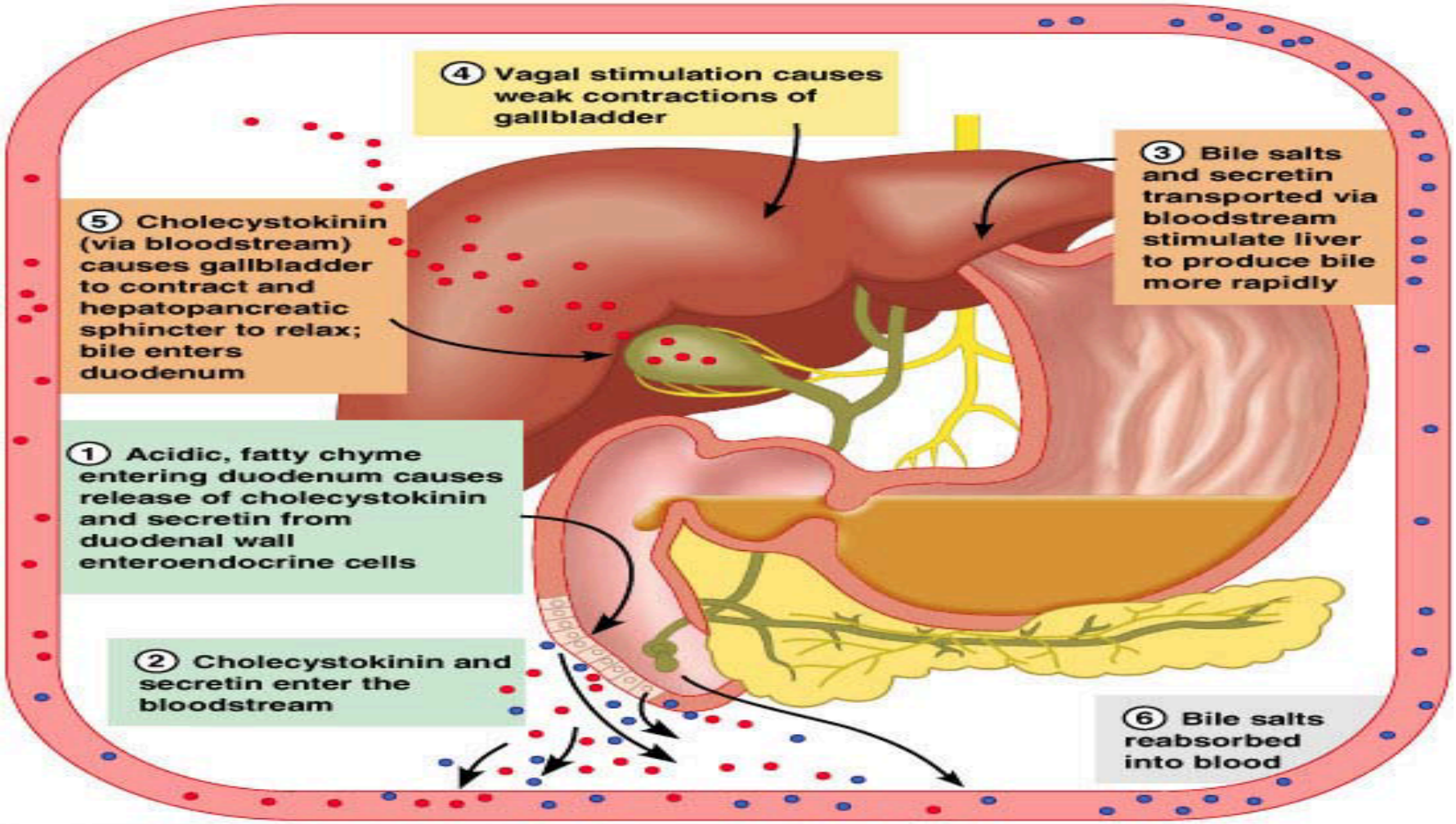


Figure: bile release into duodenum

- **Breakdown/detoxification**:
  - ✔ Ammonia to urea
  - ✔ Insulin & other hormones
  - ✔ Conjugate bilirubin
  - ✔ Breaks down toxins (drugs)
  - ✔ Breakdowns metabolites of toxic substances
  - ✔ Metabolize breakdown alcohol

➤ Metabolism/ Detoxification of Alcohol by liver

- ✔ Ethanol is part of the human diet for long
- ✔ Its excess consumption can result in a number of health problems (liver damage).
- ✔ Ethanol is metabolized, primarily by the liver

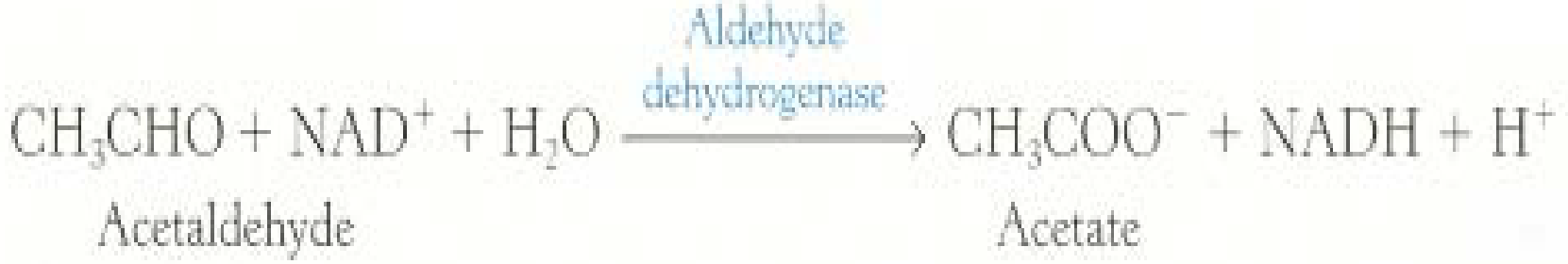
by two pathways viz:

1- The f irst pathway has two steps, the f irst step, catalyzed by the enzyme *alcohol dehydrogenase*, takes place in the cytoplasm:

-



- The second step, catalyzed by *aldehyde dehydrogenase*, takes place in mitochondria



- ✔ *These reactions leads to an accumulation of NADH*
- ✔ high [NADH] inhibits gluconeogenesis by preventing the oxidation of lactate to pyruvate
- ✔ The reverse reaction predominate, and lactate accumulates



Lactate + NADH Pyruvate + NAD

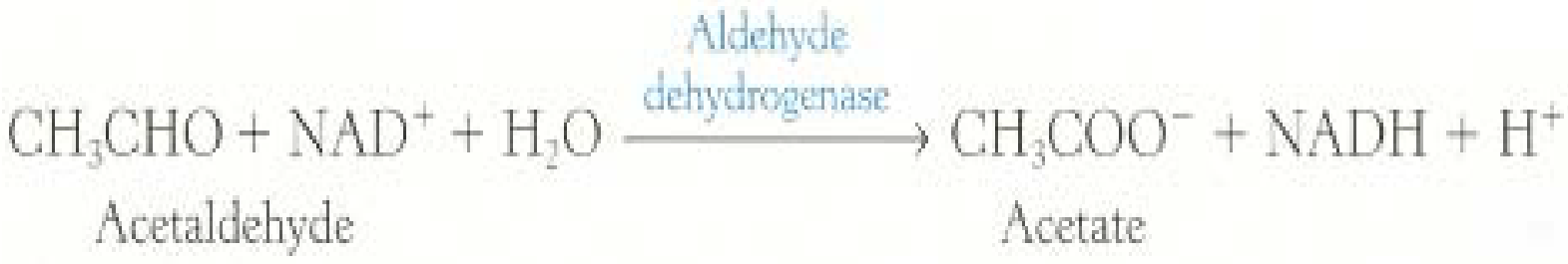
- ✔ The consequences may be hypoglycemia and lactic acidosis.

- ✔ The NADH also inhibits fatty acid oxidation
- ✔ Because NADH needs by alcohol consumer's are met by ethanol metabolism
- ✔ the excess NADH signals that conditions are right for fatty acid synthesis
- ✔ Hence, triacylglycerols accumulate in the liver, leading to a condition known as “fatty liver.”

2. The second pathway for ethanol metabolism is called

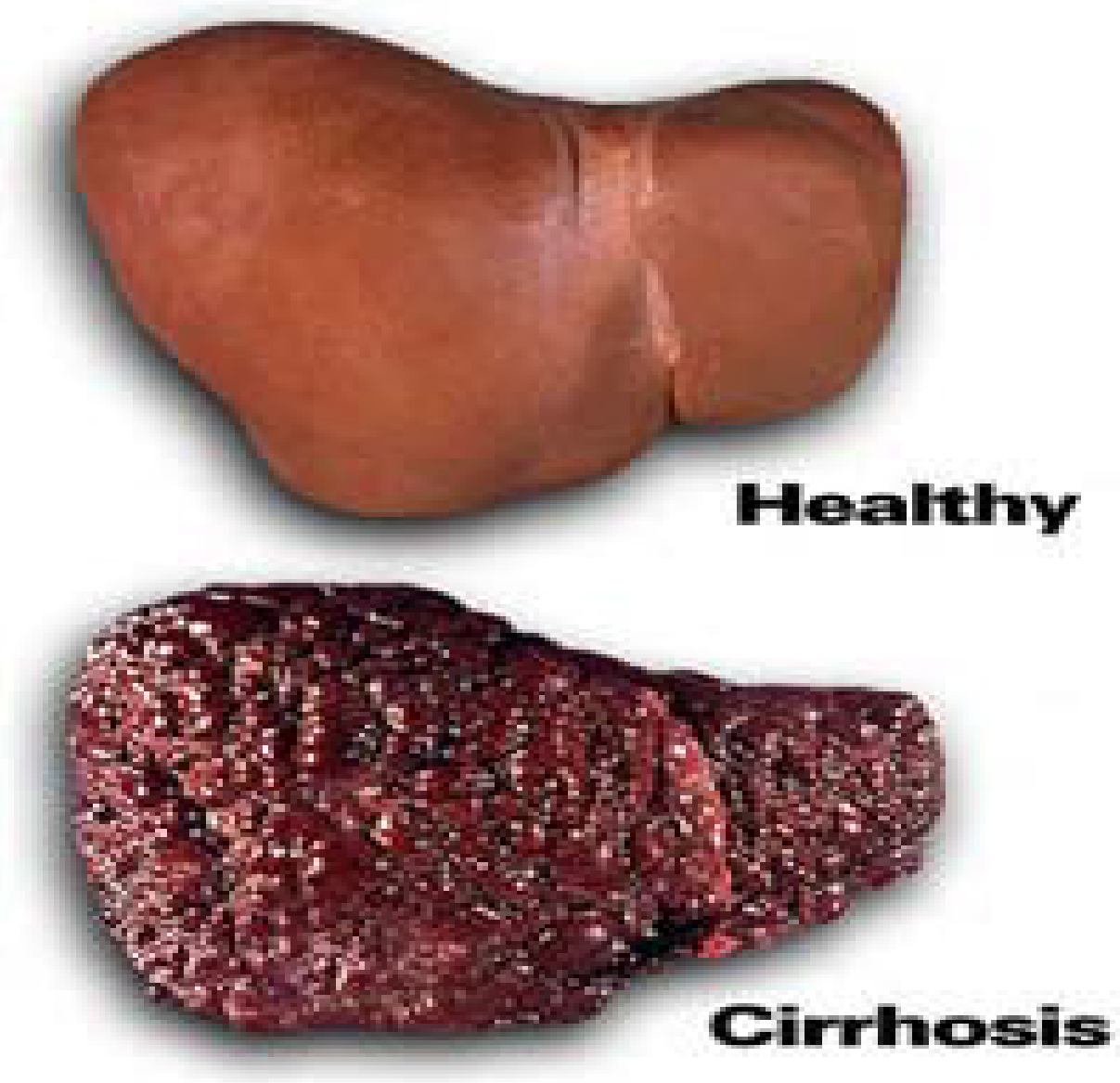
- ✔ the ethanol inducible *microsomal ethanol-oxidizing system*(MEOS).
- ✔ This cytochromeP450-dependent pathway generates acetaldehyde and subsequently acetate with concomitant NADPH, to $NADP^+$
- ✔ The pathway generates free radicals that damage tissues due to oxygen utilization.
- ✔ the antioxidant glutathione cannot be regenerated exacerbating the oxidative stress, because the system consumes NADPH,

- ✔Liver mitochondria can convert acetate into acetyl CoA via a reaction requiring ATP. The enzyme is the thiokinase that normally activates short-chain fatty acids.



- ✔ However, further processing of the acetyl CoA by the citric acid cycle is blocked, because NADH inhibits two important regulatory enzymes:
  - o Isocitrate dehydrogenase
  - o α-ketoglutarate dehydrogenase

- ✔ The accumulation of acetyl CoA has several consequences viz:-
  - o Ketone bodies will form and be released into the blood, exacerbating the acidic condition already resulting from the high lactate concentration.
  - o The processing of the acetate in the liver becomes inef fcient, leading to a buildup of acetaldehyde.
- ▪ This very reactive compound forms covalent bonds with many important functional groups in proteins, impairing protein function.
  - o ON consistently HIGH ETHANOL consumption, the acetaldehyde can signif icantly damage the liver, eventually leading to cell death.


Healthy
Cirrhosis

- Liver damage due to alcohol consumption occur in 3 stages viz: -
  - ✔ The first stage is the development of fatty liver.
  - ✔ the second stage—alcoholic hepatitis—groups of cells die and inflammation results. This stage can itself be fatal.
  - ✔ In stage three—cirrhosis—fibrous structure and scar tissue are produced around the dead cells.
  - o Cirrhosis impairs many of the liver's biochemical functions.
  - o Cirrhotic liver is unable to convert ammonia into urea, and blood levels of ammonia rise.

- Ammonia is toxic to the nervous system and can cause coma and death.
- Cirrhosis of the liver arises in about 25% of alcoholics, and about 75% of all cases of liver cirrhosis are the result of alcoholism.
- Viral hepatitis is a nonalcoholic cause of liver cirrhosis.

➤ Jaundice

Jaundice is a yellowish discoloration of the skin and of the whites of the eyes caused by abnormally high levels of the pigment bilirubinin the bloodstream.

